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INTERNATIONAL STANDARD

Superconductivity –
Part 22-1: Superconducting electronic devices – Generic specification for sensors and detectors

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SUPERCONDUCTIVITY –

Part 22-1: Superconducting electronic devices – Generic specification for sensors and detectors

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International Standard IEC 61788-22-1 has been prepared by IEC technical committee 90: Superconductivity.

The text of this standard is based on the following documents:

FDIS	Report on voting
90/388/FDIS	90/391/RVD

Full information on the voting for the approval of this International Standard can be found in the report on voting indicated in the above table.

This document has been drafted in accordance with the ISO/IEC Directives, Part 2.

A list of all parts in the IEC 61788 series, published under the general title *Superconductivity*, can be found on the IEC website.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC website under "http://webstore.iec.ch" in the data related to the specific document. At this date, the document will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

INTRODUCTION

Superconductivity offers various possibilities for the realization of sensing and detection of a variety of measurands. Several sensors and detectors have been developed, exploiting features like superconducting energy gaps, sharp normal-superconducting transition, nonlinear I–V characteristics, superconducting coherent states, and quantization of magnetic flux. All these properties can be influenced by the interaction with electromagnetic fields, photons, ions, etc. Superconducting sensors and detectors have extremely high performance for energy resolution, time response, and low noise, most of which cannot be realized by any other phenomena.

The word "sensor" is normally used for measuring stationary or slowly changing electromagnetic fields, physical quantities such as current and temperature. On the other hand, the word "detector" is normally used for single quanta such as photons from infrared to γ-rays and individual particles. However, the boundary between "sensor" and "detector" is ambiguous. In this document, therefore, both "sensor" and "detector" are used. Additionally, a detector using a sensor is possible, for example, X-ray detector using transition edge sensor (TES) that measures temperature rise due to the deposition of measurand energy. In this document, for example, the terminology "transition edge sensor X-ray detector" is used for X-ray detection using TES.

Superconducting sensors and detectors have been applied to a variety of fields including medical diagnosis, telecommunications, mineral exploration, astronomical instruments, quantum information processing, and analytical instruments. For users, IEC standardization is necessary because there is confusing terminology, there are no graphical symbols for diagrams, and no test methods.

SUPERCONDUCTIVITY –

Part 22-1: Superconducting electronic devices – Generic specification for sensors and detectors

1 Scope

This part of IEC 61788-22-1 describes general items concerning the specifications for superconducting sensors and detectors, which are the basis for specifications given in other parts of IEC 61788 for various types of sensors and detectors. The sensors and detectors described are basically made of superconducting materials and depend on superconducting phenomena or related phenomena. The objects to be measured (measurands) include magnetic fields, electromagnetic waves, photons of various energies, electrons, ions, α-particles, and others.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60027 (all parts), *Letter symbols to be used in electrical technology*

IEC 60050-815, *International Electrotechnical Vocabulary – Part 815: Superconductivity*

IEC 60417, *Graphical symbols for use on equipment* (available at: http://www.graphical-symbols.info)

IEC 60617, *Graphical symbols for diagrams* (available at: http://std.iec.ch/iec60617)

ISO 1000, *SI units and recommendations for the use of their multiples and of certain other units*

ISO 7000, *Graphical symbols for use on equipment – Registered symbols* (available at: http://www.graphical-symbols.info)

3 Terms and definitions

For the purposes of this document, the terms and definitions given in IEC 60050-815 and the following apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at http://www.electropedia.org/
- ISO Online browsing platform: available at http://www.iso.org/obp

3.1
additional positive feedback
APF
method enhancing voltage-flux transformation ratio by using resistance and coupling coil to SQUID ring

3.2
critical current modulation parameter
β_L
parameter defined by $2LI_\mathrm{c}/\Phi_0$, where L is the SQUID washer inductance, I_c is the critical current of a Josephson junction, for DC SQUIDs, and a parameter defined by $2\pi LI_\mathrm{c}/\Phi_0$ for RF SQUIDs

Note 1 to entry: The term "shielding parameter" can also be used.

3.3
Stewart-McCumber parameter
β_C
parameter defined by $2\pi I_\mathrm{c}R_\mathrm{n}^2C/\Phi_0$, where R_n is the normal state resistance of a Josephson junction, and C is the capacitance of a Josephson junction

3.4
bridge junction
junction formed from two superconductors connected by a superconducting bridge of small section

Note 1 to entry: The term "microbridge" can also be used.

3.5
critical current
I_c
maximum direct current that can be regarded as flowing through a Josephson junction without resistance

3.6
critical current density
J_c
critical current divided by the cross-section of the conductor or the junction area of the Josephson junction

3.7
feedback coil
coil that is inductively coupled to a SQUID operated in the flux locked loop (FLL) mode

3.8
flux locked loop
FLL
method that improves linearity and dynamic range of a SQUID by using negative feedback to keep the constant flux number in the SQUID ring

3.9
gradiometer
configuration of superconducting loops coupled to a SQUID magnetometer, or of multiple SQUID magnetometers that are arranged so as to be insensitive to homogenous magnetic fields or to be sensitive to magnetic field gradients

3.10
metallic magnetic calorimetric detector
type of superconducting device that the temperature increase of a metallic absorber is measured by sensing the magnetization change of the absorber because of measurand energy deposition

3.11
microwave kinetic inductance detector
type of superconducting device that uses the microwave surface impedance change of a superconducting strip because of measurand energy deposition

3.12
normal state resistance
resistance of a superconductor or a Josephson junction at a normal state

Note 1 to entry: In a superconductor or a TES, it is the resistance at a temperature just above the superconducting transition.

Note 2 to entry: In a Josephson junction, it is the tunnelling resistance at a bias voltage well above $2\Delta/e$.

3.13
planar gradiometer
kind of configuration of the flux pickup loop that measures a magnetic field gradient in the plane of the flux pickup loop coupled to a SQUID

3.14
quasiparticle
excitation combining properties of an electron and a hole that is created by breaking a Cooper pair in a superconductor

3.15
superconducting hot electron bolometric mixer
type of superconducting device that uses heterodyne mixing with the resistive transition from the superconducting state to the normal state in a superconducting microbridge because of measurand energy deposition

3.16
superconducting quantum interference device sensor
SQUID sensor
type of superconducting device that uses quantum interference occurring in a closed electrical circuit containing one or more Josephson junctions

Note 1 to entry: Every measurand, for example magnetic fields or electric currents, that is transformed into a change of magnetic flux threading the superconducting structure can be sensed by a SQUID.

3.17
SQUID array
SQA
device consisting of series and/or parallel arrays of multiple SQUIDs

3.18
nano-SQUID
device whose largest loop dimension is less than 500 nm

3.19
SQUID ring
multiply superconducting structure that contains one or more Josephson junctions

Note 1 to entry: The term "SQUID loop" can also be used.

3.20
SQUID amplifier
current–voltage converter using a single SQUID, SQUID array or other SQUID-based current sensor circuits

3.21
subgap region
lower branch of the hysteretic I–V characteristic of a tunnel junction where the voltage is less than 2Δ

3.22
subgap current
quasiparticle tunnelling current in the subgap region of a tunnel junction

Note 1 to entry: In Josephson tunnel junctions, the whole subgap region is observable when the DC Josephson effect is suppressed by applying a magnetic field parallel to the plane of the junction area.

3.23
superconducting strip detector
type of superconducting device that uses the local resistive change in a long superconducting strip because of measurand energy deposition

Note 1 to entry: The name for a photon detector, superconducting nanowire photon detector, is not recommended for most cases, since the dimensions of superconductors are often in discord with the current definition of "nanowire" in ISO/TS 80004-2:2015, in which "nanowire" or "nanofibre" is defined as nano-objects with two external dimensions in the nanoscale that are approximately 1 nm to 100 nm, and the third dimensions significantly larger. The dimensions of the superconducting strip type meet the definition of "nanoribbon" or "nanotape" in most cases.

Note 2 to entry: The terms "nanoribbon" and "nanotape" have one external dimension in the nanoscale and the other two external dimensions significantly larger (typically by more than 3 times). In addition, the two larger dimensions significantly differ from each other. "Nanostrip" is preferable to "nanoribbon" and "nanotape" for superconducting sensors and detectors.

Note 3 to entry: An example of the superconducting nanowires is the strip with the dimensions of 10 nm × 30 nm × 10 μm. The difference between the thickness and width is approximately less than 3 times. That superconductor can be called "superconducting nanowire photon detector".

3.24
superconducting tunnel junction detector
type of superconducting device that uses the change of electron tunnelling between two superconductors or a superconductor and a normal conductor separated by tunnelling barrier because of measurand energy deposition

3.25
temperature sensitivity
superconducting transition edge steepness that is defined by $\mathrm{dln}R/\mathrm{dln}T$ where R is the resistance and T is the temperature of TES

3.26
transition edge sensor detector
type of superconducting device that uses the resistive change within a sharp normal-to-superconducting transition as a temperature sensor because of measurand energy deposition

4 Symbols

Units, graphical and letter symbols shall be taken from the following standards:

- IEC 60027 (all parts);
- IEC 60417;
- IEC 60617;
- ISO 1000;
- ISO 7000.

Graphical symbols for use on equipment and diagrams, such as superconducting region, normal connection, superconducting connection, normal-superconducting boundary, Josephson junction, are defined in Annex C, and IEC 60417 and IEC 60617. Graphical symbols specific to other sensors or detectors are defined in other parts of IEC 61788.

5 Terminology and classification

5.1 Terminology

Table 1 lists the measurands which are defined as categories, objects, or physical quantities that induce enegy deposition and are to be sensed or detected by superconducting sensors and detectors. The measurands, arranged in alphabetical order, are: atoms and molecules, elementary particles, physical quantities, and radiations. Each entry in Table 1 not only represents the measurand itself, but also its temporal or spatial distribution.

Any other terminology peculiar to one of the devices covered by this document shall be taken from the relevant IEC or ISO standards.

Table 1 – Measurands

Category	Object	Physical quantity
Atoms and molecules	Atoms	Count, energy, flux, time
	Organic molecules	Count, energy, flux, time
	Nonorganic molecules	Count, energy, flux, time
	Other (specify)	
Elementary particles	Dark matters	Count, energy, flux, time
	Electrons	Count, energy, flux, time
	Neutrinos	Count, energy, flux, time
	Neutrons	Count, energy, flux, time
	Photons	Count, energy, flux, time
	Protons	Count, energy, flux, time
	Positrons	Count, energy, flux, time
	Other (specify)	
Physical quantities	Capacitance	Amplitude
	Current	Amplitude
	Inductance	Amplitude
	Magnetic field	Strength, distribution
	Magnetic flux	Density, distribution
	Magnetic susceptibility	Amplitude
	Polarization	Amplitude
	Resistance	Amplitude
	Voltage	Amplitude
	Other (specify)	
Radiations	Alpha-particles	Count, energy, flux
	Beta-particles	Count, energy, flux
	Electromagnetic waves	Amplitude
	Gamma-rays	Count, energy, flux
	Optical radiation	Count, energy, flux
	X-rays	Count, energy, flux
	Other (specify)	

The objects for measurands fall into two classes: fields and physical quantities, and particles, as listed in Table 2. Based on these classes, detection mechanisms are classified into coherent detection for fields and physical quantities, and direct detection for particles.

Table 2 – Classification of measurands

Classification	Measurand category
Field and physical quantities	Physical quantities
	Radiations (electromagnetic waves)
Particles	Atoms and molecules
	Elementary particles
	Radiations (individual electromagnetic radiation quanta)

Table 3 lists the various types of sensors and detectors (alphabetical order). The sensors and detectors convert measurands to electronic signals. The word "sensor" tends to be used for devices that measure fields and physical quantities, while the word "detector" tends to be used for devices that measure single particles. When naming sensors or detectors, the following word order should be used for nomenclature: device structure or function; measurand; and a word of detector, magnetometer, mixer, sensor, or other words. Examples of full names and corresponding acronyms are also listed.

Table 3 – Nomenclature of superconducting sensors and detectors: type, full names, and acronym examples

Type	Full names and acronym example
Metallic Magnetic Calorimetric (**MMC**) type	Metallic Magnetic Calorimetric α-ray Detector (**MMC** α-ray detector or **MMCAD**)
	Metallic Magnetic Calorimetric γ-ray Detector (**MMC** γ-ray detector or **MMCGD**)
	Metallic Magnetic Calorimetric X-ray Detector (**MMC** X-ray detector or **MMCXD**)
Microwave Kinetic Inductance (**MKI**) type	Microwave Kinetic Inductance Photon Detector (**MKI** photon detector or **MKIPD**)
	Microwave Kinetic Inductance X-ray Detector (**MKI** X-ray detector or **MKIXD**)
Superconducting Hot Electron Bolometric (**SHEB**) type	Superconducting Hot Electron Bolometric Photon Detector (**SHED** photon detector or **SHEBPD**)
	Superconducting Hot Electron Bolometric Terahertz Mixer (**SHEB** terahertz mixer or **SHEBTM**)
Superconducting Quantum Interference Device (**SQUID**) type	Superconducting Quantum Interference Device Amplifier (**SQUID** amplifier or **SQUIDA**)
	Superconducting Quantum Interference Device Current Sensor (**SQUID** current sensor or **SQUIDCS**)
	Superconducting Quantum Interference Device Gradiometer (**SQUID** gradiometer or **SQUIDG**)
	Superconducting Quantum Interference Device Magnetometer (**SQUID** magnetometer or **SQUIDM**)
	Superconducting Quantum Interference Filter Magnetometer (**SQIF** magnetometer of **SQIFM**)
	Superconducting Quantum Interference Device Array Magnetometer (**SQUID** array magnetometer or **SQUIDAM**)
Superconducting Strip (**SS**) type	Superconducting Strip Electron Detector (**SS** electron detector or **SSED**)
	Superconducting Strip Ion Detector (**SS** ion detector or **SSID**)
	Superconducting Strip Particle Detector (**SS** particle detector or **SSPD**)
	Superconducting Strip Photon Detector (**SS** photon detector or **SSPD**)

Type	Full names and acronym example
	Superconducting NanoStrip Photon Detector (**SNS** photon detector of **SNSPD**)
Superconducting Tunnel Junction (**STJ**) type	Superconducting Tunnel Junction Ion Detector (**STJ** ion detector or **STJID**)
	Superconducting Tunnel Junction Terahertz Mixer (**STJ** terahertz mixer or **STJTM**)
	Superconducting Tunnel Junction Photon Detector (**STJ** photon detector or **STJPD**)
	Superconducting Tunnel Junction X-ray Detector (**STJ** X-ray detector or **STJXD**)
	Superconductor Insulator Superconductor Terahertz Mixer (**SIS** terahertz mixer or **SISTM** that is equivalent to **STJTM**)
	Superconductor Normal-conductor Superconductor Mixer (**SNS** mixer or **SNSM**)
Transition Edge Sensor (**TES**) type	Transition Edge Sensor α-ray Detector (**TES** α-ray detector or **TESAD**)
	Transition Edge Sensor γ-ray Detector (**TES** γ-ray detector or **TESGD**)
	Transition Edge Sensor Photon Detector (**TES** photon detector or **TESPD**)
	Transition Edge Sensor X-ray detector (**TES** X-ray detector or **TESXD**)
Other	Specify

NOTE The nomenclature has the word order of (device structure or function) – (a measurand) – (a word of detector, magnetometer, mixer, sensor, or other words). The term "nanowire" can be used only for a special case that superconducting strips have two external dimensions in the nanoscale that is approximately 1 nm to 100 nm; the difference between the first and second external dimensions is typically less than 3 times; and the third dimension significantly larger: for example a nanowire with 10 nm × 30 nm × 1 µm. In IEC 61788-22, it is called "nanostrip" or "strip" because of a difference of considerably larger than 3 times in most cases, although nanotape or nanoribbon is possible. See also the ISO Online browsing platform: available at https://www.iso.org/obp/ui#home.

5.2 Classification

The operation of the sensors and detectors is classified into two categories: coherent detection and direct detection (Table 4). The devices used in the category of coherent detection include bolometers, sensors or mixers, while the devices used in the category of direct detection include calorimeters or detectors. For operating principles, see Annex A (coherent detection) and Annex B (direct detection).

Table 4 – Classification of detection principles

Detection principle	Type
Coherent detection for fields and physical quantities	SHEB, SQUID, STJ
Direct detection for particles	MMC, MKI, SS, STJ, TES

It should be noted that direct detectors have a sensitivity for single particles, and they can also measure a flux of particles. The issue is a difference between time response of superconducting devices and time interval of energy absorption of individual particles. For the superconducting sensors and detectors that use temperature rise because of measurand energy absorption, they are called "bolometer" or "calorimeter" depending on the time response. When the interval of absorption events is much shorter than the time response, the term "bolometer" is appropriate for flux measurement. On the other hand, when it is much longer than the time response, the term "calorimeter" is appropriate for particle counting.

6 Cryogenic operation condition

The superconducting sensors and detectors operate in cryogenic temperatures lower than T_c or the middle of normal-superconducting transition in order to hold a superconducting state or to use a sharp normal-superconducting transition for sensing or detecting measurands. The operation temperature range depends on the types. Specific cryogenic operation condition for each sensor or detector should be described in other parts of IEC 61788-22 for various types of sensors and detectors.

The cryogenic temperature can be obtained by liquid helium or other cryogens, Gifford-McMahon (GM) coolers, pulse tube coolers, adiabatic demagnetization refrigerators (ADR), ^{3}He refrigerators, ^{4}He-^{3}He dilution refrigerators, and other cryocoolers.

Performance of sensors and detectors is influenced by both cryogenic environment and implementation. The cryogenic environment and implementation should be specified in a standard for each sensor or detector and other separate standards.

7 Marking

7.1 Device identification

The marking on the device or on the packing shall enable clear identification of the device. The device shall be provided with a traceability code which enables back-tracing of the device to a certain production or inspection lot.

7.2 Packing

Marking on the packing shall state the following:

a) the device identification code;
b) the measurand, type, name, detection principle, and other informative information in Tables 1, 2, 3, and 4;
c) the number of enclosed devices;
d) the operating temperature range;
e) the additional specification;
f) the required precautions, if any.

8 Test and measurement procedures

The test and measurement methods should be described in a standard for each sensor or detector and other separate standards.

Annex A
(informative)

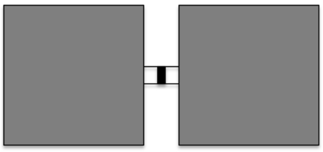
Coherent detection

A.1 Superconducting hot electron bolometric (SHEB) type

The SHEB type enables detection of sub-millimetre and far-IR radiation (0,3 THz to 10 THz). SHEBs are more often used as indirect (i.e. coherent or mixing) detectors, although direct detection is possible. In mixers, the electromagnetic field of the signal photons (ω_s) is mixed with a locally generated electromagnetic field at ω_0 frequency (called the local oscillator, LO). The result is a signal at the difference, or intermediate frequency, ω_{IF}, where $\omega_{IF} = \omega_0 - \omega_s$. The original signal in the sub-millimetre band is therefore down-converted into the microwave band, where more conventional low-noise electronics can amplify and process the signal. A SHEB is composed by a superconducting microbridge with nanometre or submicron dimensions, contacted by thick metallic pads. Sub-millimetre radiation signals are coupled into the microbridge through a lens and an on-chip antenna. The heterodyne mixing process makes use of the resistive transition between the superconducting state and the normal state of the superconducting bridge, induced by the heating of radiation signals. This transition produces a hot-spot. Two types of hot-spot relaxation are used as a detection mechanism: in the diffusion cooled SHEB the hot electrons diffuse to the surrounding thermal reservoir, while in the phonon cooled SHEB, the hot electrons are cooled through fast electron–phonon interaction.

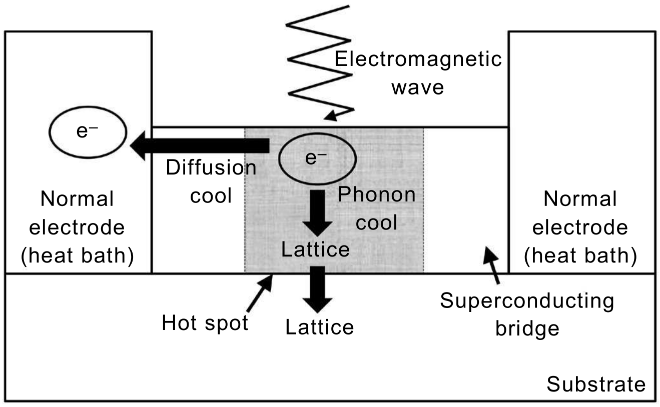
Bolometers are electronic devices that convert measurands into heat, which can then be detected by a thermometer. Although conventional bolometers usually have antenna-shaped absorber, thermometer, heat sink, and thermal link as separate elements, in the SHEB type these various elements are combined. The thermal resistance $R = 1/G$ with the thermal conductance G provides the coefficient linking incident power P and change in temperature ΔT, such that $\Delta T = PR$. An important consideration in designing bolometers is the time taken for it to recover, given by $\tau = RC$, where C is the heat capacity. In SHEB, at a low temperature, the electron system in materials becomes decoupled from the phonon system, leading to a large R, while C of the electrons becomes small and proportional to T.

From the physics of device operation, the SHEB type can be categorized as a kind of TES. In general TES, temperature of both lattice and electron increase due to the incident photon or electromagnetic wave, while only the electron temperature does for SHEB type. Since the heat capacity C of the electron is not bigger than C of the lattice except for extremely low temperature, SHEB type can respond more quickly to the incident signal than general TES.



Two pads are connected by microbridge. The size of normal conducting hot spot region changes depending on absorbed energy.

a) SHEB type mixer concept



The signal and local oscillator powers and joule heating due to DC bias produce a hot spot whose length oscillates at $|\omega_0 - \omega_s|$ in the superconducting bridge.

b) Schematic of a hot electron bolometer

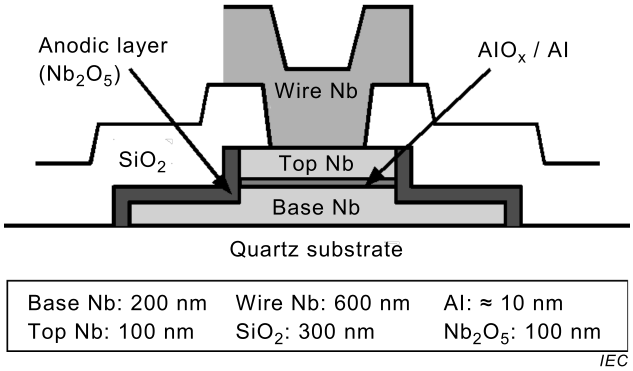
Figure A.1 – SHEB mixer

An example of the SHEB type is a device that consists of two thick metal pads that are connected or bridged by a small superconducting microbridge of NbN with a thickness of 3,5 nm, a width of 400 nm, and a length of 4 µm (see Figure A.1 a) and A.1 b). Other material examples are Nb, NbTiN, Al, and $YBa_2Cu_3O_x$.

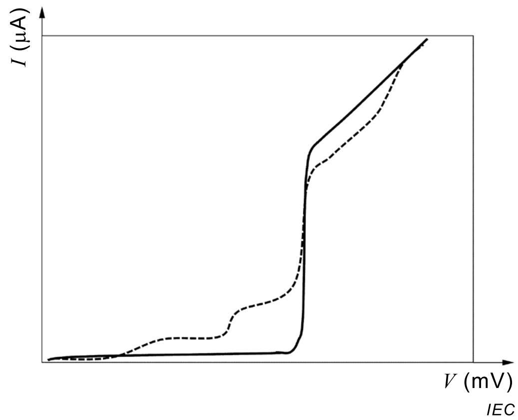
An advantage of the SHEB type is that it operates at very high speeds through either fast phonon or electron diffusion cooling, which enables heterodyne detection of radiation with a frequency above about 1 000 GHz that the STJ mixers cannot receive efficiently because of quasiparticle excitation above a superconducting energy gap. SHEBs can also be used for direct detection, but their main application is heterodyne detection.

A.2 Superconducting tunnel junction (STJ) type

The STJ structure is exactly the same as that of Josephson tunnel junctions (Figure A.2 a)), but the Josephson effect needs to be suppressed by a magnetic field applied in parallel to the junction surface. The STJ type enables coherent detection based on heterodyne mixing of measurand of electromagnetic waves in millimetre- and submillimetre-wave bands and a local oscillator frequency in order to downconvert frequencies into the microwave band by using strong non-linear I–V curves (Figure A.2 b)). The STJ mixers are frequently called superconductor insulator superconductor (SIS) mixers due to their sandwiched structure. STJ mixers operating at a cryogenic temperature (e.g. 4 K) are indispensable for frequencies less than about 1 000 GHz, in which there are no low-noise amplifiers. At the downconverted frequencies that are normally in the microwave frequency band, low-noise semiconductor-based amplifiers are available.



a) Typical structure of a Nb/Al/AlO_x/Nb STJ



b) I–V curves: without local oscillator power (solid line) and with local oscillator power (dashed line)

Figure A.2 – STJ mixer

An example of STJ is a Nb/Al/AlO_x/Nb junction, shown in Figure A.2 b). The size and critical current density decreases and increases, respectively, as the signal frequency increases so as to keep the normal resistance approximately constant. The typical size and critical current density of a junction at 500 GHz (0,6 mm in wavelength) are approximately 1 μm and 100×10^6 A/m^2 at 4,2K, respectively.

An advantage of STJ mixers is very high spectral resolution compared with direct detection of photons, although STJ mixers have a disadvantage in sensitivity.

A.3 Superconducting quantum interference device (SQUID) type

The SQUID type enables coherent detection of the quantum state of a superconducting loop containing one Josephson junction (RF SQUID) or two Josephson junctions (DC SQUID (Figure A.3), based on magnetic flux quantization and the Josephson effect. The coherent state is influenced by magnetic flux coupled to the SQUID loop. By means of this flux coupling, any measurand that can be transformed into magnetic flux threading the SQUID loop can be measured by a SQUID sensor. Prominent examples are extremely sensitive SQUID magnetometers or current sensors. In addition, magnetic susceptibility and magnetic moment of materials can also be measured with SQUID sensors. Shielding current or penetration of magnetic flux quanta (Φ_0 = 2,067 8 $\times$ 10^{-15} Wb) into the superconducting SQUID loop results in periodic oscillations of the voltage vs. magnetic flux or the current vs. magnetic flux characteristics of SQUIDs. The period of these oscillations is one flux quantum Φ_0. SQUID sensors can, however, have sensitivities better than one Φ_0 and can be used with or without a feedback circuit as a null detector of magnetic flux.

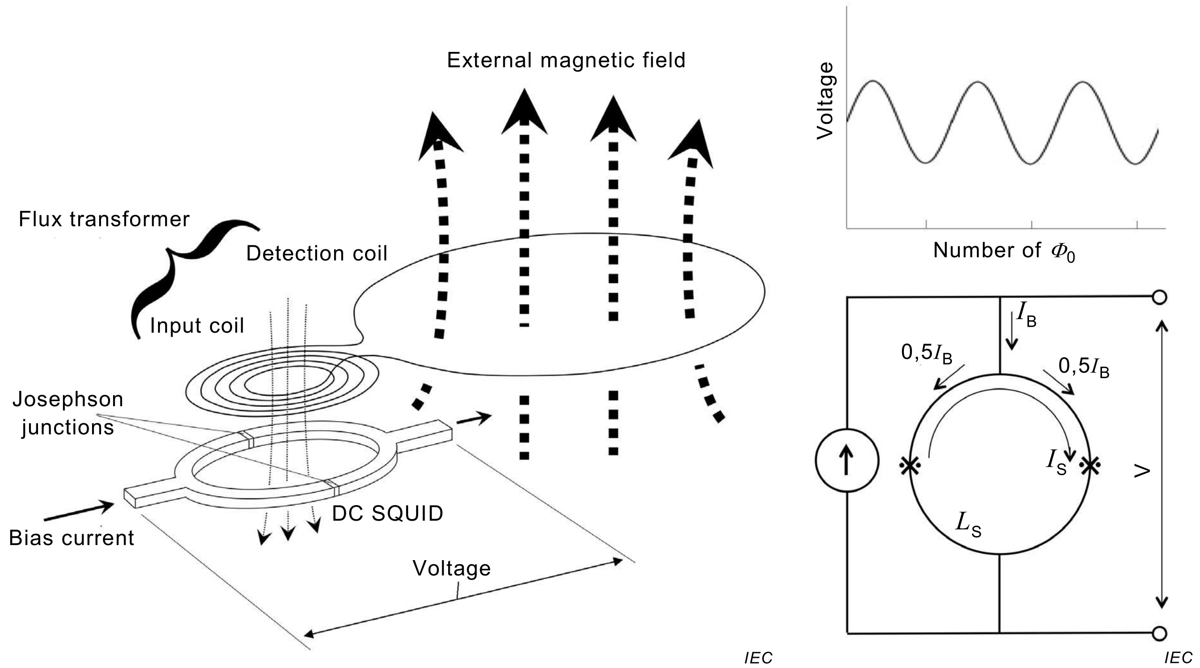
A low T_c SQUID made of niobium-based alloy is typically operated at 4,2 K (e.g. in liquid helium) and a high T_c SQUID is operated at 77 K (in liquid nitrogen). Fundamentally, operation principles of both SQUIDs are the same but the sensitivities are different. The noise level of the low T_c SQUID is the order of 1 fT/Hz$^{1/2}$ and that of high T_c is 10 times larger.

Figure A.3 a) is a schematic illustration of a typical magnetometer using a DC SQUID. A flux transformer consisting of a large detection coil and a spiral input coil is used. An external magnetic field is transformed into magnetic flux threading the SQUID loop by the transformer. A Φ_0-periodic voltage vs. magnetic flux response is obtained (Figure A.3 b)).

A simplified equivalent circuit of a DC SQUID is shown in Figure A.3 b). A single DC SQUID is usually operated in a constant current-biased condition. The bias current (I_B) flows and splits into the two paths, each with currents 1/2 I_B. The two Josephson junctions are located in the superconducting loop. There is a normal resistance R associated with each Josephson junction. If the density of the magnetic flux threading the DC SQUID loop is the same as the density of the magnetic flux out of the loop, nothing happens. But if there is imbalance of the each density, the shielding current I_s is induced and flows in the loop to compensate the imbalance. I_s is expressed by $I_s = \varphi/L_s$ (φ: total flux threading the SQUID loop, L_s: SQUID loop inductance).

The shielding current I_s is in the same direction as I_B in one of the branches of the superconducting loop, and is opposite to I_B in the other branch; the total current becomes $(I_B + I_s)/2$ in one branch and $(I_B - I_s)/2$ in the other. As soon as the current in either branch exceeds the critical current I_c of the Josephson junction, a voltage appears across the junction.

The value of the voltage across the junction periodically changes along with the flux in the loop. If the feedback technology FLL (Flux Locked Loop) is applied, the FLL output voltage is proportional to the change of the flux out of the SQUID loop.



a) SQUID type sensor concept: a magnetometer using a superconducting detection coil coupled to a DC SQUID via a flux transformer

b) Circuit diagram of a DC SQUID and the voltage vs. magnetic flux characteristics

Figure A.3 – DC SQUID

An example of SQUID is a Nb superconducting loop having two Josephson junctions of $Nb/AlO_x/Al/Nb$ with a junction size of 1 µm, a critical current density of 25×10^6 A/m^2 at 4,2 K, and a junction normal resistance R of 8 Ω. Frequently, a spiral input coil that generates magnetic fields is integrated on the SQUID chip.

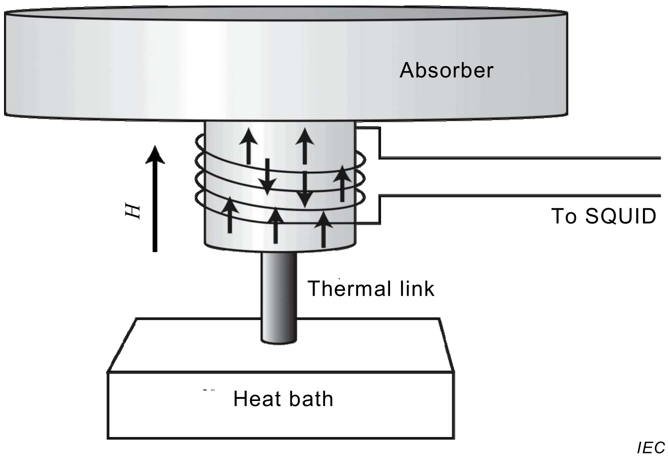
Fundamentally, SQUIDs are sensitive to magnetic flux. However, the use of input coils and flux transformers allows for the measurement of others quantities such as magnetic field and current.

Annex B
(informative)

Direct detection

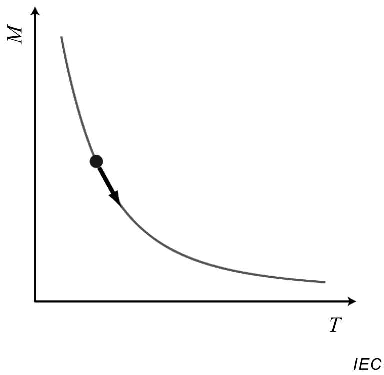
B.1 Metallic magnetic calorimetric (MMC) type

The MMC type enables direct detection based on the measurement of the change of magnetization of a paramagnetic sensor in a small magnetic field upon the deposition of measurand energy. The MMC type consists of a paramagnetic sensor whose magnetization changes upon temperature rise and a superconducting circuit that reads the magnetization change. The energy of each measurand (e.g. a photon) causes a temperature increase of an absorber attached to the paramagnetic sensor. The temperature rise results in a magnetic susceptibility change (or decrease in magnetization as shown in Figure B.1) of the absorber that is measured by the magnetometer under a small magnetic field. The absorber is thermally isolated from a low temperature thermal bath, so that the magnetization change is large enough compared with a noise level of the magnetometer.



An absorber is connected to a paramagnetic material, whose magnetization changes with increasing temperature.

a) MMC detector concept



Energy absorption of measurand in an absorber eventually results in a magnetization decrease that is measured by a SQUID.

b) Temperature dependence of paramagnetic sensor material (e.g. Au:Er)

Key

H magnetic field strength

M magnetization

T temperature

Figure B.1 – MMC detector

An example of the paramagnetic sensor is Au:Er paramagnetic metallic alloy that is a small concentration of erbium, 300 ppm[1] to 1 000 ppm, in gold host. The paramagnetic sensor and the absorber are placed in a small magnetic field at a cryogenic temperature, typically 1 mT to 10 mT, and 25 mK. The temperature increase of the gold absorber results in a magnetization decrease as shown in Figure B.1 b). This change is read by a superconducting circuit using a SQUID. The response time of the spin system in Au:Er is fast (less than 1 μs at 100 mK) due to strong coupling between the magnetic spins and conduction electrons in the metal. The

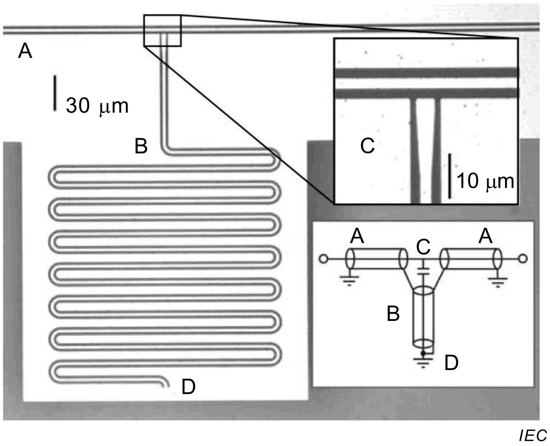
1 ppm = parts per million

decay time of MMC signals depends upon the thermal link of the detector to a thermal bath at low temperatures.

An advantage of MMC detectors is that the energy resolution (ΔE) depends weakly on the heat capacity (C) of the absorber. It is expected that $\Delta E \propto C^{1/3}$, which means that magnetic calorimeters are suited for applications for which a large size absorber is needed.

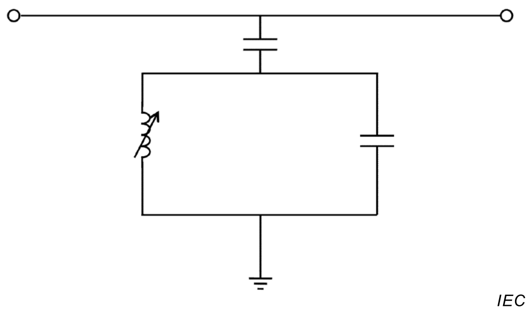
B.2 Microwave kinetic inductance (MKI) type

The MKI type enables direct detection based on the measurement of the surface impedance change of a superconducting film through the kinetic inductance effect upon the deposition of measurand energy. The surface impedance change is dominated by the change of kinetic inductance of the superconductor. A microwave kinetic inductance detector (MKID) consists of electric components that form a resonator that is excited at a resonance frequency in a microwave frequency range (Figure B.2). The resonance curve changes when the energy of a measurand (e.g. a photon) is deposited in the superconducting inductor through the kinetic inductance change due to Cooper pair breaking. If the resonator is excited with the resonance frequency, the absorbed energy is determined by measuring the degree of phase change and the resonance amplitude change because of the change of resistance (Figure B.2 b)).



A coplanar waveguide is connected to a meander resonator.

a) MKI type detector concept



Surface resistance change due to measurand energy deposition results in the changes of phase and resonance amplitude. By preparing different resonance frequencies for a large number of pixels, it is possible to distinguish the signal of each pixel.

b) Circuit diagram

Key

A coplanar wave guide

B meander resonator

C ground plane

D ground

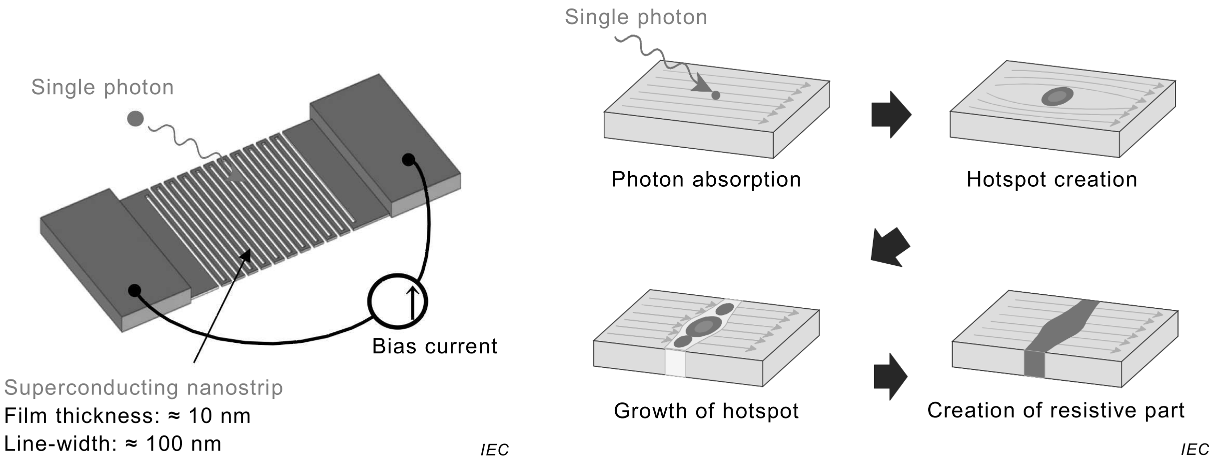
Figure B.2 – MKI detector

An example of the absorber is a TiN_x strip with a thickness of 60 nm and a length of 3 mm at 0,1 K. The TiN_x strip forms a meander line inductor that is coupled with a resonator capacitor. The resonator is connected to a coplanar waveguide used for resonator excitation by microwave in the frequency range from 0,1 GHz to 20 GHz and read out through a coupling capacitance (Figure B.2). The excitation and the measurement of the phase change and the resonance amplitude change are controlled by a sophisticated microwave component at room temperature.

An advantage of MKIDs is that large scale arrays are possible. The resonance frequencies of the resonators (pixels) can be tuned at a certain interval without overlap of the resonance curves of individual resonators. Thousands of pixels can be read out over a single microwave cable that is laid between a cryogenic temperature and room temperature.

B.3 Superconducting strip (SS) type

The SS type enables direct detection based on the measurement of resistive state occurrence induced by hot spot formation or other physical phenomena in a superconducting strip fabricated on a substrate after the deposition of measurand energy. Bias current is applied below the critical current values of the strip. This detector is a type of superconducting transition edge device. When the hot spot size is large enough so that current density in the surrounding area exceeds the critical current density, a resistive region appears across the width of the strip (Figure B.3). The resistive band relaxes and thermal energy inside dissipates into the substrate within a time considerably less than 1 ns.



a) SS type detector concept **b) Operating principle of SS type detector**

Figure B.3 – SS detector

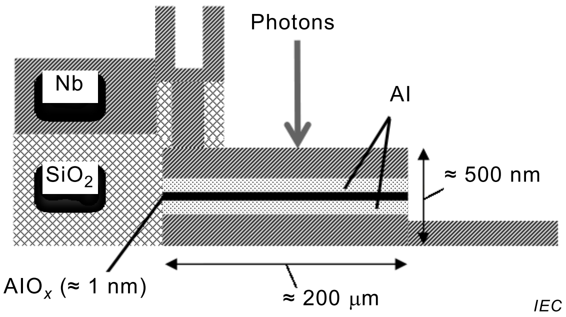
An example of SS is a NbN strip with a thickness of 5 nm and a width of 100 nm. The NbN strip is cooled at 4 K. In order to obtain a reasonable detector size, strips are formed in a meander shape or in parallel lines.

An advantage of SS is a very fast response time (better than a few nanoseconds) compared with other sensors and detectors. The time response is mainly determined by kinetic inductance of strips, since intrinsic relaxation time of excitation in superconductors is much shorter than kinetic inductance limited time response. On the other hand, spectral resolution is unexpected or very poor.

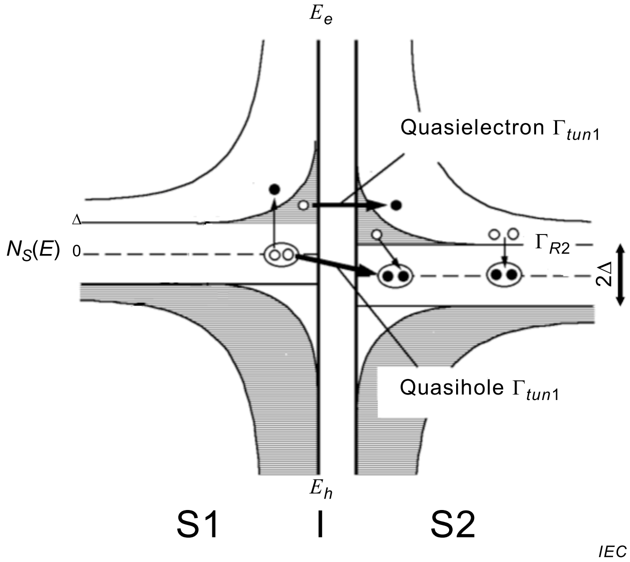
B.4 Superconducting tunnel junction (STJ) type

The STJ type enables also direct detection based on quasiparticle excitation induced by measurand absorption. STJ has a structure that is exactly the same as that of Josephson junctions, but the Josephson effect needs to be suppressed by a magnetic field applied in parallel to the junction surface (Figure B.4 a)). The junction is biased within a subgap region that is less than $2\Delta/e$ and cooled considerably below T_c in order to avoid quasiparticle creation by thermal excitation. When either superconducting electrode absorbs the energy of any measurand, the Cooper pairs break and excess quasiparticles are created. The excess quasiparticles tunnelling through the thin insulator (I) between two superconducting electrodes (S1 and S2) result in the increase of tunnelling current across the junction (Figure

B.4 b)). The number of quasiparticles is proportional to E/Δ. This current returns to zero according to quasiparticle lifetime.



a) **STJ type detector concept**



Two main tunnelling processes of quasiparticles for signal creation and recombination into Cooper pairs are shown.

b) **Operating principle of STJ type detector**

Key

Δ superconducting energy gap

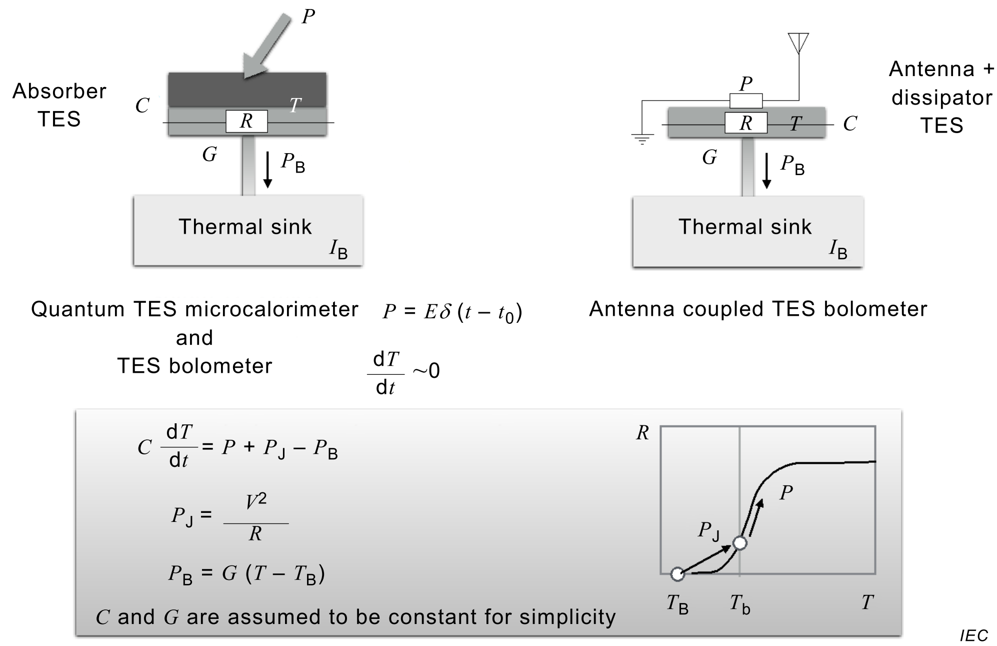
Figure B.4 – STJ detector

An example of STJ is a Nb/Al/AlO_x/Al/Nb junction with a junction size of 200 µm, a critical current density of 2×10^6 A/m^2 at 4,2 K, and a normal resistance of 20 Ω. The junction size is considerable larger than that of SIS mixers, because a substantially large sensitive area is required.

An advantage of STJ is the superconducting energy gap that ensures high stability against base temperature drift, when it is kept about one-tenth of T_c of the superconducting electrodes.

B.5 Transition edge sensor (TES) type

The TES type enables direct detection based on resistive change induced by equilibrium temperature change due to the deposition of measurand energy in the middle of the normal-superconducting transition of a film (Figure B.5). The temperature of TES is kept at a temperature in the middle of the normal-superconducting transition either by adjusting the temperature precisely or applying a constant voltage V across both ends of the film at a temperature somewhat lower than T_c of the superconducting film. In the latter case, the self-heating power of $P_J = V^2/R$ balances with the heat escape from the film to a heat bath, P_B. The R value of the film increases with increasing temperature within the normal-superconducting transition, which results in the decrease of P_J. Therefore, the temperature of the film is stabilized at a bias point, T_b, in the transition edge depending on the V value. Accordingly, for example, when a photon is absorbed in an absorber that is attached to TES causes a temperature rise, the self-heating power decreases. This provides negative feedback to make the TES return to the bias point more quickly than the thermal time constant. The effect is called electrothermal feedback (ETF) (Figure B.5). The negative current pulse upon the measurand absorption is read out by a SQUID current amplifier.



TES type detector concept and operating principle of TES type detector

Key

C heat capacity of absorber

E power of measurand

G thermal conductance between absorber and thermal sink

P external heat power due to measurand energy deposition

P_B thermal flow through a connection between absorber and thermal sink

P_J joule heating power due to bias voltage and resistance of transition edge sensor

R resistance of transition edge sensor

T temperature

T_B base temperature

T_b operating temperature of transition edge sensor

t time

V bias voltage

Figure B.5 – TES detector

An example of TES is a bilayer of Mo(60 nm)/Cu(200 nm) with an area of 400 µm × 400 µm on a Si_3N_4 membrane with a thickness of 250 nm at 70 mK. The TES is shunted by a Cu resistor of 0,2 mΩ that is considerably smaller than an R value at the bias position, which provides constant bias voltage. The negative current pulses for individual absorption events of measurand energy are converted into voltage pulses by a series SQUID array amplifier.

An advantage of TES is a combination of extremely high energy resolution of calorimeter performance and stable biasing in the middle of the narrow normal-superconducting transition width, of which the steep slope of R-T curve provides a highly sensitive temperature sensor. Furthermore, ETF can compensate for an inherent slow response of calorimeters and shorten pulse relaxation time.

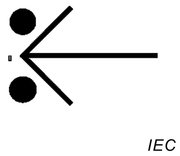
In this document, films for TES are defined as superconducting films, since they are influenced by superconductivity in spite of finite resistance. The devices that use the influence of superconductivity are called superconducting devices. Energy resolution depends on C.

Annex C
(normative)

Graphical symbols for use on equipment and diagrams

C.1 Superconducting region, one superconducting connection

See Figure C.1.

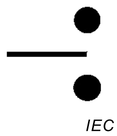


The vertical two dots pair is the superconducting region and the perpendicular line with the right angle wedge is the superconducting connection.

Figure C.1 – Superconducting region, one superconducting connection

C.2 Superconducting region, one normal-conducting connection

See Figure C.2.

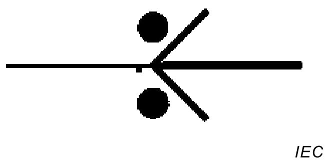


The vertical two dots pair is the superconducting region and the perpendicular line is the normal-conducting connection.

Figure C.2 – Superconducting region, one normal-conducting connection

C.3 Normal-superconducting boundary

See Figure C.3.



The vertical two dots pair is the superconducting region, the perpendicular line with the right angle wedge on the right hand side is the superconducting connection, and the perpendicular line on the left hand side is the normal-conducting connection.

Figure C.3 – Superconducting region, one superconducting connection, and one normal-conducting connection (normal-superconducting boundary, IEC 60417-6370:2016-09)

C.4 A variation

See Figure C.4.

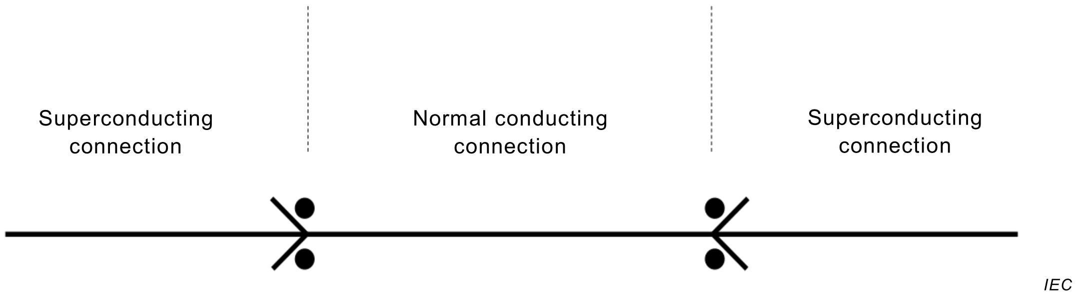
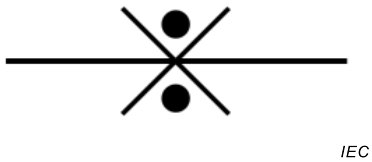


Figure C.4 – Series connection

C.5 Josephson junction

See Figure C.5.



The vertical two dots pair is the superconducting region, the perpendicular lines with the right angle wedges are the superconducting connections from both sides. The non-superconducting region which is normal conducting or insulating between two superconducting connections is extremely small.

Figure C.5 – Superconducting region, two superconducting connections with extremely small non-superconducting region (Josephson junction, IEC 60417-6371:2016-09)

The symbol of Figure C.3 and C.5 are registered with IEC 60417 Graphical symbols for use on equipment. The registration with IEC 60617 Graphical symbols for diagrams is in progress.

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